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CS 362

AI Reflection

- Awad distinguishes between different “types” of AI. What classification scheme does the paper use and why do these types matter for scientific research?

The way Awad classifies the different types of AI is listing them as paradigms. These different paradigms include various models of AI each with their own uses that benefits the user. These models include, descriptive AI, predictive AI, generative AI, and more. These matter for scientific research as they can be optimal solutions for specific problems.

- Does Awad make a clear distinction between AI as a tool and AI as a scientific collaborator? If so, what are the differences and what are some examples given to support the differences? Do these examples suggest a real shift in how science is conducted, or mostly an extension of existing methods?

I believe Awad does make a clear distinction between AI as a tool and as a scientific collaborator. Towards the end of describing how each model works, Awad gives an example on what would be the best use case for each model. For some, the model can be used as tools for tasks that can be done by humans but may take a long time such as looking through large data sets. There are others that can be considered collaborators to science, like **Generative Adversarial Networks (GANs)** from the generative AI models. In this case, the model can gather data and simulate real-world results that could explore innovative ideas and help with new hypotheses or research.

- What are some limitations or risks of using AI in science? How do these relate to issues such as interpretability, bias, reproducibility, or theory formation?

Some limitations or risks of using AI in science lean more towards the ethics of how models are used and the results it gives. For example, with **Natural Language Processing** it is possible for it to produce results with unintended bias that contain race and stereotypical stereotypes. This effects the credibility of the scientific work if it were to be published without proper review

- According to Awad’s arguments, is AI more likely to accelerate scientific discovery or to reshape the scientific method itself? Do you agree or disagree?

- This is a tricky one to think about. For now, I would agree with Awad that AI is more likely to accelerate scientific discovery. There were many examples throughout the text that demonstrated how different AI models can speed up the process of work done by scientists. This can open new doors for ideas that arise and want to be tested at a faster pace rather than spending months going through data to simulate the hypothesis. To say that AI is reshaping the scientific method itself can be tricky if the models are trained to follow the current method. Perhaps later models can figure out more optimal methods for different fields to conduct research.